QAIS MOMANI

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Summary

First year CDT Student in Robotics and AI for Net Zero with experience in autonomous robotics, robotic manipulation, ROS 2, embedded systems and multidisciplinary system integration. Interested in researching field robotics, autonomous inspection and the deployment of robotic systems in challenging real-world environments.

Education

MEng (Hons) Mechatronic Engineering: First Class Honours **06/2026**
The University of Manchester **Manchester, UK**

Experience

PhD Researcher: Robotics and AI for Net Zero (RAINZ) **09/2026 to current**
The University of Manchester **Manchester, UK**

- Undertaking an MSc in Robotics as part of the RAINZ CDT programme, developing advanced knowledge of autonomous robotic systems, perception, manipulation and systems integration.
- Research project focused on developing robotic inspection capabilities for confined steel and concrete structures, with applications in challenging environments.

Nanotechnology Research Intern **06/2024 to 08/2024**
The University of Manchester **Manchester, UK**

- Increased nanopore production efficiency by building a bespoke MATLAB GUI that integrated a Keithley 2614b SMU for precise voltage control, real-time data visualisation, and a critical safety cut-off mechanism.
- Authored a 22-page technical report, synthesising findings from over 15 academic papers on nanopore and Controlled Dielectric Breakdown theory to contribute to future research efforts.

Skills

Programming: Python, C, MATLAB, C++.
Professional Development: IET Graduate Member, Machine Learning Course (MIT Open Learning Library).
Software: ROS2, Linux, SolidWorks, KiCAD, Simulink.
Languages: Arabic and English (fluent), French (intermediate).

Projects

Autonomous Litter Picking Robotic System for Urban Environments- The University of Manchester

- Led development of the litter retrieval subsystem for SAPLiNG, a two-robot autonomous litter picking system sponsored by Amentum, using an AgileX PiPER 6-DOF arm, RGB-D perception, and 2 AgileX Scout Minis.
- Integrated manipulation with detection by developing a custom geometric grasp pose generation algorithm that converts litter point cloud data into feasible end effector poses.
- Integrated the retrieval ROS 2 architecture, including a Python coordination node, grasp pose service interface, and C++ MoveIt2 service nodes for arm motion and gripper control.
- Validated retrieval performance with 97% success in isolated arm and gripper trials, 72% success with autonomous grasp generation, and 100% pick-up in clutter testing when multiple attempts were allowed.
- Managed the power architecture of the system, including an external power supply for the arm and camera, DC-DC conversion, wiring, and integration with the Scout Mini platform.

Introducing Amyloid-Related Imaging Abnormalities (ARIA) to the Brain- The University of Manchester

- Investigated the clinical presentation and MRI characteristics of Amyloid-Related Imaging Abnormalities, utilising 39 academic sources to establish the theoretical framework for developing anatomically accurate lesion models.
- Engineered a scalable MATLAB program to generate synthetic 3D ARIA lesions using Gaussian models and anatomical tissue masks, executing in under 2 seconds with negligible memory increase (<4 kB).
- Composed an 8,240-word dissertation that addressed the critical shortage of MRI data for training AI detection algorithms, documented the project's methodology and validated the models against clinical MRI data.

Embedded Systems Line-Following Buggy Project- The University of Manchester

- Led the electromechanical design, performing DC motor characterisation to select an optimal gear ratio of 10.8:1 and engineering a multi-layer chassis using SolidWorks to integrate 6 core components and simplify wiring.
- Resolved critical hardware faults under a tight deadline, including repairing a motor's spline shaft and using an oscilloscope to identify a faulty H-bridge circuit in the motor drive board, restoring full power for the demonstrations.